

Project Title: Binary Lock With Three Attempts

Group Members:

S. No.	Name	Reg #
1	Muhammad Maaz Mushtaq	Cs251041
2	Shavez Yousaf	Cs251075
3	Abdul Rahim Shera	Cs251306

1. Project Introduction:

- **Overview:** This project presents the design and implementation of a secure digital locking system utilizing fundamental Digital Logic Design (DLD) principles. The system functions as an electronic security gateway that permits access only upon the entry of a predefined 4-bit binary password ("1001").
- **Motivation and Purpose:** The primary motivation for choosing this project is to demonstrate the practical application of basic logic gates and sequential components in real-world security scenarios. The goal was to bridge the gap between theoretical circuit design and functional hardware implementation, specifically addressing the challenge of preventing unauthorized access through brute-force attempts.
- **Expected Outcomes:** The primary outcome of this project is a highly reliable, 100% hardware-based digital security lock that operates entirely on digital logic, without requiring any microcontrollers or software programming. By successfully integrating combinational logic (for exact password validation) and sequential logic (for attempt tracking and memory), the system achieves a robust and self-sustaining security protocol. The successful implementation of the transistor-based auto-reset ensures a seamless user experience, while the auto-lock and active alarm system guarantee a fail-safe mechanism against continuous brute-force attacks.
- **Real-World Applications:** The fundamental logic implemented in this project serves as the backbone for numerous modern security systems. Key real-world applications include:
 - **Banking & ATM Security:** Automated Teller Machines (ATMs) utilize this exact "3-attempt limit" logic. If a user enters an incorrect PIN three times consecutively, the system automatically locks the interface or captures the card to prevent fraudulent access.
 - **Smart Home & Office Access Control:** Digital door locks, electronic safes, and keypad entry systems in high-security facilities rely on this hardware logic to grant access or sound an alarm upon tampering.
 - **Hardware Security Modules (HSM):** Industrial vaults and critical storage units prefer hardware-based auto-locking mechanisms—which cannot be easily bypassed or "hacked" like software—to freeze access immediately upon unauthorized entry attempts.

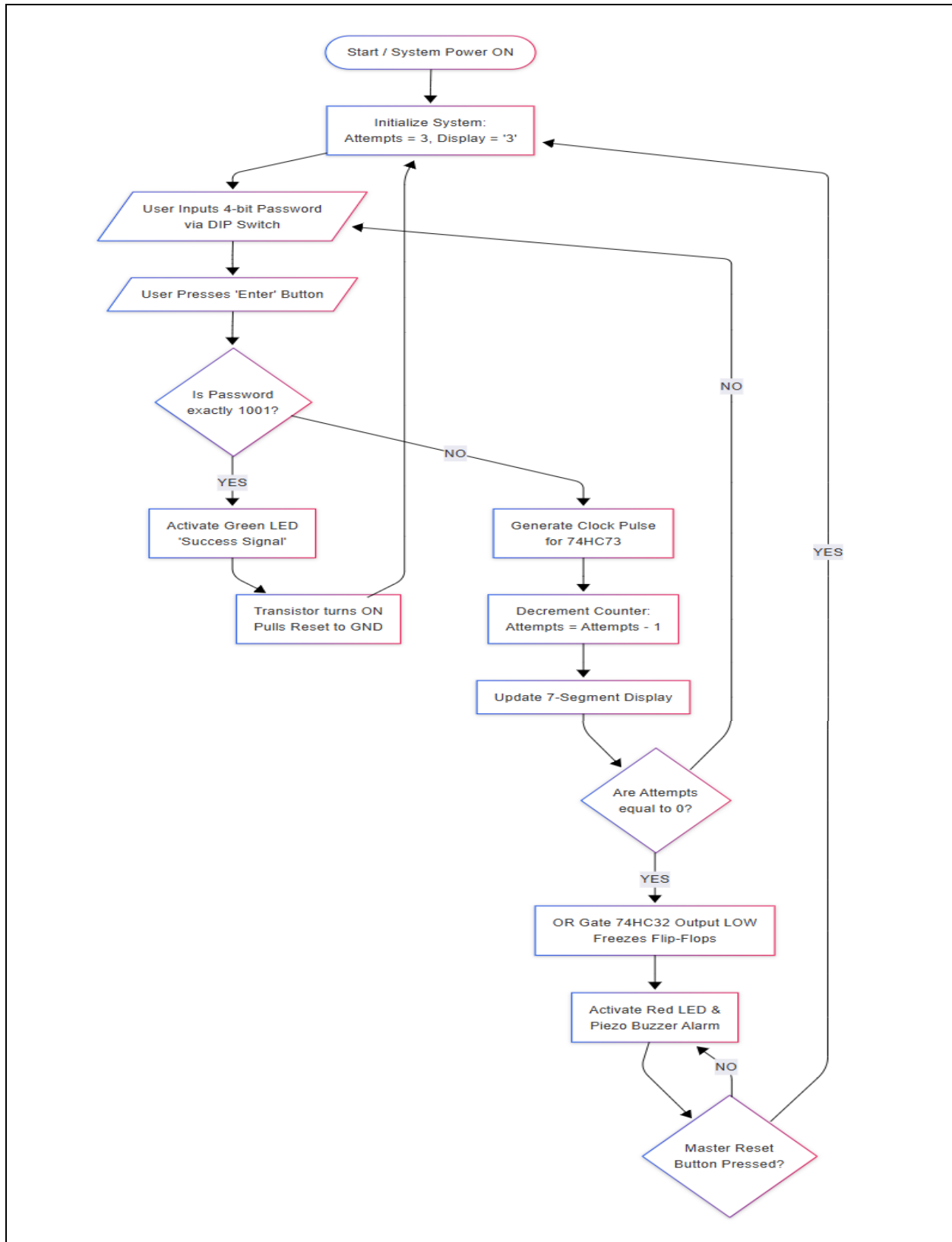
- **Electronic Device Lockouts:** The core concept of disabling a smartphone or a laptop for a specific duration after multiple wrong password attempts originates directly from the digital counting and locking logic demonstrated in this project.

2. Working Principle

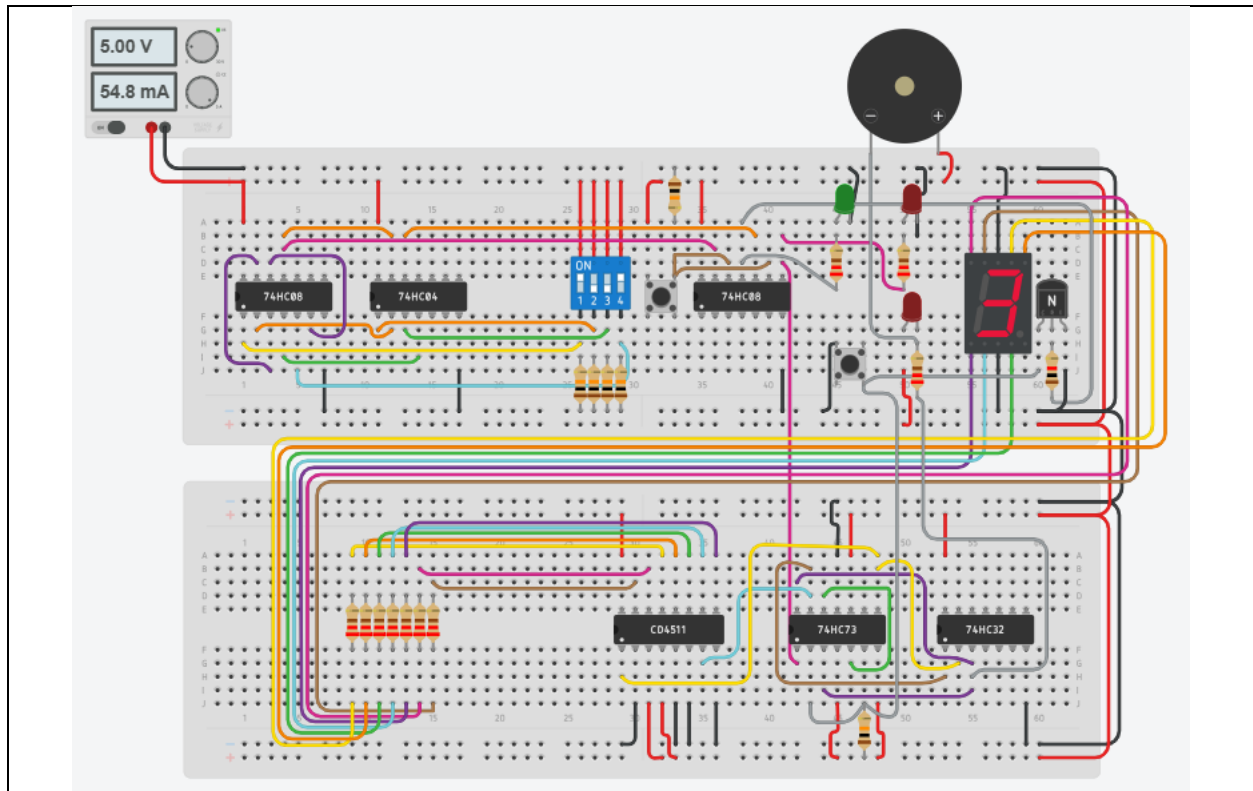
The project operates on a hybrid architecture combining both Combinational and Sequential Logic:

1. **Combinational Logic (Authentication):** A 4-position DIP switch acts as the input mechanism. The predefined password "1001" is verified using 74HC04 (NOT) gates to invert the '0' inputs and 74HC08 (AND) gates to validate the exact bit sequence. A pushbutton triggers the verification process. If the input matches, a HIGH signal activates the Green "Success" LED.
2. **Sequential Logic (Attempt Tracking):** A 74HC73 Dual J-K Flip-Flop is utilized as a counter to track failed attempts. It is triggered by the clock pulse generated from an incorrect password entry. The inverted outputs of the flip-flops are fed into a CD4511 BCD-to-7-Segment Decoder to visually display a countdown from 3 to 0.
3. **Auto-Lock Mechanism:** Upon reaching 0 attempts, a 74HC32 (OR) gate detects the LOW states and effectively freezes the flip-flop's J and K inputs, halting any further counting. Simultaneously, it creates a current sinking path to activate a Red LED and a 5V Piezo Buzzer to signify system lockdown.
4. **Hardware Optimization (Auto-Reset):** To enhance user experience, an NPN Transistor (BC547) acts as an active-low electronic switch. Upon a correct password entry, the transistor receives a HIGH signal at its base, bridging the collector and emitter to pull the Clear pins of the 74HC73 to GND. This automatically resets the attempt counter back to '3' without manual intervention.

3. Block Diagram / Logic Diagram



4. Screenshot (TinkerCAD / CircuitWizard)



5. Results / Conclusion

- **Conclusion:** The project was successfully designed and simulated. The system flawlessly executed its core functionalities: verifying the correct "1001" password, decrementing the attempts counter exclusively on incorrect inputs, displaying remaining attempts dynamically, and triggering an automated lockdown and alarm at zero attempts. Furthermore, the integration of the transistor-based reset proved highly effective in automating the system's recovery phase upon a successful login.
- **Learning Outcomes:** Designing this project provided profound insights into the practical implementation of DLD concepts. We successfully applied combinational logic for secure data validation and utilized sequential logic (J-K Flip-Flops) for state memory and counting. Moreover, we learned critical hardware interfacing techniques, such as using current sinking for active-low alarms, pulling down inputs to prevent floating states, and utilizing NPN transistors as electronic switches to bypass logical constraints (diode limitations) in reset circuits.

6. References

- Digital Logic Design (DLD) Lab Manual, DHA Suffa University.
- "Digital Design" textbook by M. Morris Mano (for logic gates and flip-flop concepts).
- Tinkercad (for circuit simulation and testing) - www.tinkercad.com
- Official Datasheets for 74HC series ICs (74HC08, 74HC73, 74HC32) and CD4511 from www.alldatasheet.com.
- Online DLD tutorials and logic gate reference materials from ElectronicsHub.